SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Total Marks:25**

Annexure A

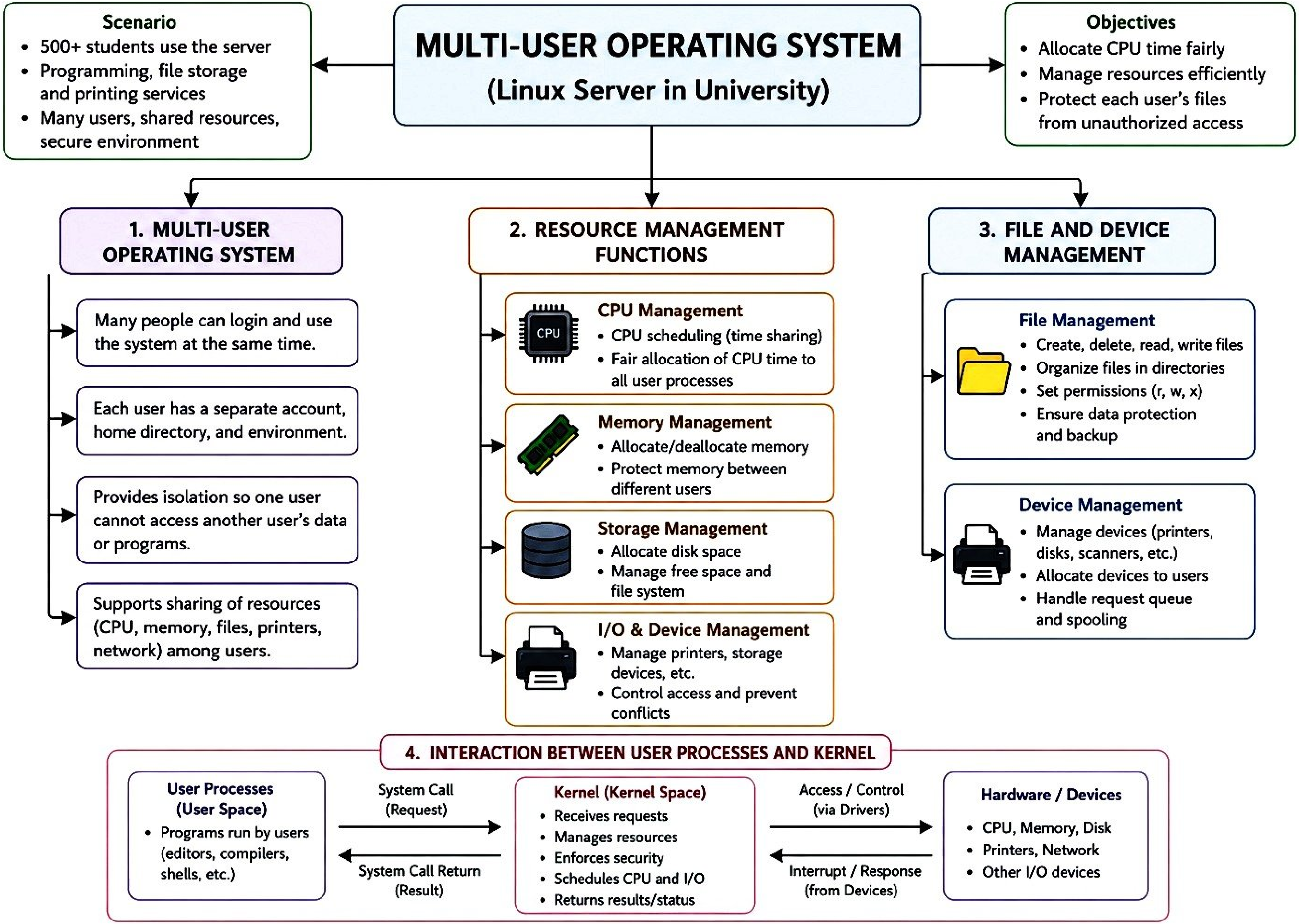
Concept Mapping

Code of Conduct:

I Atchaya Chandran R(192521394) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

R. Atchaya


MULTI-USER OPERATING SYSTEM
(Linux Server in University)
Scenario
• 500+ students use the server
• Programming, file storage and printing services
• Many users, shared resources, secure environment
Objectives
• Allocate CPU time fairly
• Manage resources efficiently
• Protect each user's files from unauthorized access
1. MULTI-USER OPERATING SYSTEM
Many people can login and use the system at the same time.
Each user has a separate account, home directory, and environment.
Provides isolation so one user cannot access another user's data or programs.
Supports sharing of resources (CPU, memory, files, printers, network) among users.
2. RESOURCE MANAGEMENT FUNCTIONS
CPU
CPU Management
• CPU scheduling (time sharing)
• Fair allocation of CPU time to all user processes
Memory Management
• Allocate/deallocate memory
• Protect memory between different users
Storage Management
• Allocate disk space
• Manage free space and file system
I/O & Device Management
• Manage printers, storage devices, etc.
• Control access and prevent conflicts
3. FILE AND DEVICE MANAGEMENT
File Management
• Create, delete, read, write files
• Organize files in directories
• Set permissions (r, w, x)
• Ensure data protection and backup
Device Management
• Manage devices (printers, disks, scanners, etc.)
• Allocate devices to users
• Handle request queue and spooling
4. INTERACTION BETWEEN USER PROCESSES AND KERNEL
User Processes (User Space)
• Programs run by users (editors, compilers, shells, etc.)
System Call (Request)
System Call Return (Result)
Kernel (Kernel Space)
• Receives requests
• Manages resources
• Enforces security
• Schedules CPU and I/O
• Returns results/status
Access / Control (via Drivers)
Interrupt / Response (from Devices)
Hardware / Devices
• CPU, Memory, Disk
• Printers, Network
• Other I/O devices